

MCQs

These are fragments of what we recalled after 5 days of exam. Please go through the topics of the question rather than the questions and the options alone...

Clinical

1. A middle-aged man with epilepsy is brought to A&E with carbamazepine-induced hyponatraemia. During his time in the hospital, he is also diagnosed with depression. Which antidepressant would be the best choice for this patient?

Select one:

- A. Mirtazapine**
- B. Amitriptyline
- C. Sertraline
- D. Citalopram
- E. Venlafaxine

Mirtazapine is recommended for the treatment of depression in people with epilepsy. It is not known to be proconvulsive and it has nonsignificant pharmacokinetic drug interactions. Furthermore, it carries a low risk of inducing hyponatraemia. SSRIs may be proconvulsant in overdose. SSRIs with the lowest risk of interactions with antiseizure medications are generally preferred in the setting of epilepsy (citalopram/escitalopram, followed by sertraline). SSRIs have a relatively high risk of inducing hyponatraemia. There is mixed evidence for venlafaxine increasing the seizure risk, and it is considered as a moderate risk for people with epilepsy. Most tricyclic antidepressants are epileptogenic at higher doses (particularly clomipramine and amitriptyline) and are considered high-risk in the setting of epilepsy.

2. A 33-year-old woman with moderate intellectual disability presents with mania. Which of the following medications would you not prescribe?

Select one:

- A. Valproate**
- B. Lithium
- C. Carbamazepine
- D. Lamotrigine
- E. Levetiracetam

Valproate is an established human teratogen. NICE does NOT recommend valproate for the treatment of mania in women of child-bearing potential. Most now agree that valproate should not be used in women under 50 years of age, and an outright prohibition of its use in these patients is being considered in some countries.

3. An elderly man with Alzheimer dementia has shown agitation while in hospital. CT brain has revealed an acute posterior brain infarct. What is the most appropriate management for his condition?

Select one:

- A. Avoid the use of psychotropic drugs if possible**
- B. Start low-dose haloperidol
- C. Start low-dose risperidone
- D. Start a low-dose benzodiazepine
- E. Start low-dose olanzapine

The first steps when encountering behavioural and psychological symptoms of dementia (BPSD) are thorough assessment and management of contributing medical problems. Non-pharmacological interventions for BPSD should always be tried before pharmacological interventions. Patients with dementia often already have significant polypharmacy which may contribute to BPSD. Risperidone (and haloperidol) are the only drugs licensed in the UK for the management of non-cognitive symptoms associated with dementia. Due to the serious adverse effects of haloperidol, risperidone is the agent of choice in cases of persistent aggression in patients with moderate-to-severe AD unresponsive to non-pharmacological approaches and when there is a risk of harm to self or others. A careful risk assessment should be done, balancing the cerebrovascular risk. Given that this patient has an acute brain infarct, psychotropic drugs should be avoided if possible.

4. What is the most common behavioural and psychological symptom of dementia seen in patients with Alzheimer dementia?

Select one:

- A. Apathy**
- B. Aggression
- C. Irritability
- D. Aberrant motor behaviour
- E. Disinhibition

Commonly-observed core symptoms of dementia have been classified as behavioural and psychological symptoms of dementia (BPSD) or neuropsychiatric symptoms (NPS). BPSD/NPS are extremely common in Alzheimer dementia: apathy is the most frequent symptom, followed by depression, aggression, anxiety, and sleep disorder. Other symptoms, such as irritability, appetite disturbance, aberrant motor behaviour, delusions, disinhibition, and hallucinations are less common.

5. An older man is interested in reducing his risk of dementia. Which of the following is most likely to reduce his risk?

Select one:

- A. Quitting smoking**
- B. Taking vitamin B and zinc supplements
- C. Taking omega 3 supplements
- D. Nothing can be done to reduce his risk
- E. Taking daily aspirin

It has been proposed that a high proportion of cases of late life dementia can be attributed to cardiovascular risk factors (e.g. hypertension, diabetes, and obesity) and psychosocial factors (e.g. education, social engagement, and leisure activities). It is estimated that a third of late life dementia might be attributable to seven modifiable risk factors – low education, mid-life hypertension, mid-life obesity, diabetes, physical inactivity, smoking, and depression.

[SPMM] In large epidemiological studies, which of the following is the biggest modifiable risk factor in later life contributing to population attributable risk (PAR) for dementia?

Select one:

- A. Social isolation
- B. Alcohol misuse
- C. Physical inactivity
- D. Smoking**

E. Air pollution

EXPLANATION: A growing body of evidence supports several potentially modifiable risk factors for dementia. Risk factors in early life (less education), midlife (hypertension, obesity, hearing loss, traumatic brain injury, and alcohol misuse), and later life (smoking, depression, physical inactivity, social isolation, diabetes, and air pollution) can contribute to increased dementia risk. Of the later life risk factors, smoking has the highest population attributable fraction. Stopping smoking, even in later life, ameliorates this risk.

6. Which of the following is a key role for CBT in psychosis?

Select one:

A. Distress reduction

- B. Confrontation
- C. Validation
- D. Reattribution
- E. Acceptance

According to NICE, CBT for psychosis should be offered so that people can establish links between their thoughts, feelings, or actions and their current or past symptoms, and/or functioning and can then re-evaluate their perceptions, beliefs, or reasoning as they relate to the target symptoms. CBT can promote alternative ways of coping with target symptoms, reduce distress, and improve functioning. In the cognitive model of psychosis, coping styles are conceptualized as the behavioural response to symptoms. Responses may be helpful or maladaptive and serve to maintain problematic beliefs and potentially worsen symptoms (e.g., isolating self with voices are present, leading to increased voices and distress). Thus, modifying these coping styles can be an important tool in reducing distress.

7. Which of the following types of therapy utilises phone calls as a core technique?

Select one:

A. Dialectical behaviour therapy

- B. Cognitive behavioural therapy
- C. Psychoanalytic psychotherapy
- D. Interpersonal psychotherapy
- E. Cognitive analytic therapy

Dialectical behaviour therapy (DBT) was introduced in 1991 by Marsha Linehan and colleagues as a treatment for borderline personality disorder (BPD). Patients with BPD are supported in understanding their own emotional experiences and are taught new skills for dealing with their distress through a combination of group and individual therapy sessions. DBT is a complex treatment combining CBT interventions with Eastern meditative practice, notably mindfulness. Core techniques include individual therapy, group skills training, telephone contact, and therapist consultation groups.

8. A patient who has been prescribed antipsychotics has been non-compliant with his medication. After using amphetamines, he presents with mutism, cataplexy, and fever. What is the most likely diagnosis?

Select one:

A. Malignant catatonia

- B. Selective mutism
- C. Malignant hyperthermia
- D. Neuroleptic malignant syndrome
- E. Serotonin syndrome

Malignant catatonia, previously termed ‘lethal catatonia’, is the most severe manifestation within the spectrum of catatonic syndromes. It may include immobility, rigidity, mutism, posturing, excessive motor activity, stupor, negativism, staring, and echolalia. It can occur in the setting of an underlying neuropsychiatric syndrome or general medical disorder (e.g. drug-related, as in this case) and shares clinical and pathophysiological features and medical comorbidities with neuroleptic malignant syndrome.

9.Red flag MEED

- A. BP (cannot remember the values)
- B. Pulse 45/ min
- C. Recent loss of weight 1kg per week**
- D. Temperature 36.5
- E. BMI 13.5 *Red flag is less than 13.*

Full MEED Guidelines – Full Table

Physical observation	
BP	Under 18 → below 0.4 th centile Over 18 → less than 90 Systolic drop more than 20 mmHg or HR increase more than 30 bpm
HR	<40
Temp	35.5 or 35 axillary
Examination	
Dehydration	10% loss and fluid refusal
BMI	Under 18 → < 70% Over 18 → <13
Weight loss	>1kg / week for 2 consecutive weeks
ECG and Biochemistry	
ECG	< 18 → QTC > 460 females, 450 males > 18 → QTC > 450 females, 430 males
Biochemistry	➔ Hypophosphatemia ➔ Hypokalaemia <2.5 mmol/L ➔ Hypoalbuminemia ➔ Hypoglycaemia < 3mmol/l ➔ Hyponatremia ➔ Hypocalcaemia ➔ Transaminase more than 3x normal range ➔ DM with HbA1c > 10% (86mmol/L)
FBC	Low WBC Hb less than 10g/L

10. Age to prescribe Fluoxetine as per NICE guideline

- A. 8
- B. 4
- C. **5**
- D. 6
- E. 7

11. Clonidine is not effective for

- A. Inattention
- B. Impulsivity
- C. Hyperactivity
- D. Aggression
- E. **Overarousal**

[SPMM] Clonidine has been shown to be efficacious in the treatment of which feature of ADHD?

Select one:

- A. Hyperactivity
- B. Inattention
- C. Impulsivity
- D. Aggression
- E. **All of the listed features**

The non-stimulants atomoxetine, clonidine, and guanfacine have been shown to be efficacious in treating ADHD, but their effect sizes compared to placebo are generally in the medium range and smaller than those of stimulants. Improvements in inattention, impulsivity, and hyperactivity have all been reported with clonidine. The alpha-2 agonists (clonidine and guanfacine) may also be particularly helpful for ADHD with associated aggression and other features of conduct disorder.

12. Wife works abroad a lot, didn't care of the husband. husband died suddenly. Wife very angry and shouts at doctor. Which defence mechanism is she displaying

- A. Denial
- B. Reaction formation
- C. Sublimation
- D. Projection
- E. Splitting
- F. **Displacement**
- G. Acting out

[SPMM] Miss Baker is a 26-year-old woman who was made redundant due to issues around her work performance. She refuses to acknowledge that there is any issue with her work performance. She believes that she was sacked because one of her colleagues was gossiping about her. After her dismissal, Miss Baker posted abusive messages against this colleague on her Facebook page. Which of the following is the defence mechanism being used?

Select one:

- A. Displacement**

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- B. Projective identification
- C. Repression
- D. Denial
- E. Reaction formation

Displacement involves redirecting emotion to a safer outlet; the separation of emotion from its real object and redirection of the intense emotion toward someone or something that is less offensive or threatening in order to avoid dealing directly with what is frightening or threatening. In this case, the patient finds it threatening to accept that they may have been sacked for poor work performance but easier to blame another individual.

[Psych-Mentor] A man with a borderline personality disorder begins to cancel appointments with his therapist. He also starts engaging in increased alcohol consumption. The therapist notes that this occurs at a time when the therapeutic work has become very challenging. Which of the following defence mechanisms is suggested?

- A. Sublimation
- B. Projective identification
- C. Displacement
- D. Regression
- E. Acting out**

Missed appointments and substance misuse are common examples of acting out.

13. Patient with Parkinson's disease came with his wife. He see small animals, people in home . but he enjoys there company. Which should be the next step?

- A. Add quetiapine most probably
- B. Add nothing**
- C. Add memantine
- D. Add Risperidone
- E. Clozapine

Psychosis in PD – recommendations for treatment

Step	Intervention
1	Exclude organic causes (delirium).
2	Optimise the environment to maximise orientation and minimise problems due to poor caregiver–patient interactions.
3	If the patient has insight and hallucinations are infrequent and not troubling, do not treat.
4	Consider reducing or stopping anticholinergics and dopamine agonists. Monitor for signs of motor deterioration. Be prepared to restart/increase the dose of these drugs again to achieve the best balance between psychosis and mobility.
5	Consider an atypical antipsychotic. The efficacy of clozapine (see below, point 7) is supported by placebo-controlled RCTs.²⁹ In contrast, there are several negative placebo-controlled trials each for quetiapine and olanzapine.²⁹ Low-dose quetiapine is the best tolerated, although EPSEs and stereotypical movements can occur. It is probably reasonable to try quetiapine³⁶ before clozapine but the success rate may be low. Olanzapine, ziprasidone and aripiprazole are likely to all have greater adverse effects on motor function than quetiapine, although one small trial³⁷ supports the safe use of ziprasidone. Risperidone and typical antipsychotics should be avoided completely. Severe rebound psychosis has been described when antipsychotic drugs (quetiapine or clozapine) are discontinued. All antipsychotics may be relatively less effective in managing psychotic symptoms in patients with dementia, and such patients may be more prone to developing motor and cognitive side effects.³⁸ Antipsychotics have been associated with an increased risk of vascular events in the elderly. In PD all antipsychotics are linked to increased mortality³⁹ although the effect of clozapine is not known.
6	Consider a cholinesterase inhibitor , particularly if the patient has co-morbid dementia ^{29,40} Cholinesterase inhibitors may also reduce the risk of falls. ⁴¹ Early use of donepezil does not prevent or reduce episodes of psychosis although there is some benefit on cognition. ⁴²

(Continued)

14. A middle aged man live in a care house , he needs assistance sometimes in regular , cannot do finance , life and more descriptions, cannot cook, cannot manage own finances

A. Moderate

B. Severe id

C. Mild id

D. Profound

E. Borderline

The following table illustrates the categories of intellectual difficulties.

IQ	Category	Description
50-69	Mild intellectual disability	Approximately 85% of cases; most can lead normal lives except may need assistance in handling difficult situations
35-49	Moderate intellectual disability	Approximately 10% cases; use simple language when talking but understand speech better. Patient can generally attend to the basic tasks of life after training but more complex activities such as using money usually require support within a special residential environment
20-34	Severe intellectual disability	Approximately 3-4% cases: many able to look after themselves with careful supervision
<20	Profound intellectual disability	Approximately 1-2% cases: development level of one year old baby across a range of parameters and so require intensive help and supervision in all activities

15. What is one of “R” in CAT

A. Revision

B. Recovery

C. Role Reversal

D. Recall

E. Review

➤ **Reformulation**

➤ **Recognition**

➤ **Revision**

Stats

EMIs

Clinical

16. Which gait is classically seen in

1. Upper motor neuron lesions – **Scissoring gait**
2. Parkinson disease – Festinating gait

Select one:

- A. Scissoring gait and siffing
- B. High steppage gait
- C. Antalgic gait
- D. Ataxic gait
- E. Magnetic gait

Gait	Description	Associated conditions
Festinating (Shuffling)	Posture is stooped forward. Gait initiation is slow and steps are small and shuffling	Parkinson's disease
Ataxic	Gait is wide-based with truncal instability and irregular lurching steps which results in lateral veering and if severe, falling	Cerebellar disease, e.g. Wernicke's
Antalgic	Stance phase of gait is abnormally shortened relative to the swing phase, usually done to minimise pain	Lower limb trauma
Spastic (scissor or diplegic)	Rigidity and excessive adduction of the leg in swing, plantar flexion of the ankle, flexion at the knee, adduction and internal rotation at the hip, and contractures of all spastic muscles	Cerebral palsy
Steppage (neuropathic or equine)	High-stepping gait so as to prevent scraping of the toe on the ground	Foot drop
Myopathic (waddling)	A broad-based gait with a duck-like waddle to the swing phase, the pelvis drops to the side of the leg being raised with forward curvature of the lumbar spine, and a marked body swing	Proximal myopathy
Pigeon	Toe(s) point(s) inwards when walking	Structural abnormalities
Trendelenburg	During the stance phase, the weakened abductor muscles allow the pelvis to tilt down on the opposite side. To compensate, the trunk lurches to the weakened side to attempt to maintain a level pelvis throughout the gait cycle	Poliomyelitis or muscular dystrophy
Stomping	Bilateral high steppage due to lack of proprioception	Friedreich's ataxia
Magnetic	Feet seem as if magnetically attracted to the floor	Normal pressure hydrocephalus
Choreiform (hyperkinetic)	Irregular, jerky, involuntary movements in all extremities. Walking may accentuate their baseline movement disorder	Sydenham chorea, Huntington's
Sensory	In an effort to know when the feet land and its location, the patient will slam the foot hard onto the ground in order to sense it	proprioceptive loss e.g. diabetes, B12 deficiency

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17. Durations for diagnosis in ICD 11

1. OCD
2. GAD
3. Hypochondriasis
4. Mania

Duration of symptoms needed to make a diagnosis

According to ICD-11, what is the minimum duration of symptoms needed to make a diagnosis of each of the following?

1. Manic episode ✗
2. Generalised anxiety disorder ✗
3. Hypochondriasis ✗
4. Depressive episode ✗
5. PTSD ✗

1. Manic episode - **1 week**
2. Generalised anxiety disorder - **Several months**
3. Hypochondriasis - **Duration not specified**
4. Depressive episode - **2 weeks**
5. PTSD - **Several weeks**
6. OCD - **Duration not specified.**

[ICD11] **Obsessive-Compulsive Disorder** is characterised by the presence of persistent obsessions or compulsions, or most commonly both. Obsessions are repetitive and persistent thoughts, images, or impulses/urges that are intrusive, unwanted, and are commonly associated with anxiety. The individual attempts to ignore or suppress obsessions or to neutralize them by performing compulsions. Compulsions are repetitive behaviours including repetitive mental acts that the individual feels driven to perform in response to an obsession, according to rigid rules, or to achieve a sense of 'completeness'. In order for obsessive-compulsive disorder to be diagnosed, obsessions and compulsions must be time consuming (e.g. taking more than an hour per day) or result in significant distress or significant impairment in personal, family, social, educational, occupational or other important areas of functioning.

18. Yalom's curative factor, allowing group members speak and burst of emotion and getting new view of life, helping other, getting self esteem high __ altruism, something about group cohesion, one member s identifies other member as family

Yalom's therapeutic/curative factors in group therapy

Choose the therapeutic/curative factor described in each group therapy scenario.

1. A group member sobs as she recounts a previous traumatic experience, and then feels relieved after she finishes.

✗

2. A group member who usually only thinks about himself finds value in being able to help another member.

✗

3. Group members start to feel that they are working together toward a common goal.

✗

4. Several members share knowledge and give advice to a new member, and this impacts his subsequent choices.

✗

5. The comments of senior group members produce a sense of optimism in someone who has just recently joined the group.

✗

Catharsis
Corrective familial experience
Altruism
Group cohesion
Development of socialising techniques
Interpersonal learning
Instillation of hope
Universality

EXPLANATION:

- ➔ **Catharsis** is the expression of ideas, thoughts, and suppressed material that is accompanied by an emotional response that produces a state of relief in the patient.
- ➔ **Altruism** is the act of one member's being of help to another; putting another person's needs before one's own and learning that there is value in giving to others.
- ➔ **Group cohesion** is the sense that the group is working together toward a common goal. It is also referred to as a sense of 'we-ness'.
- ➔ **Interpersonal learning** occurs when group members acquire knowledge about new areas. They receive advice, obtain guidance, and attempt to influence and are influenced by other group members.
- ➔ **Installation of hope** (also known as 'inspiration') is the process of imparting a sense of optimism to group members; the ability to recognise that one has the capacity to overcome problems.

Therapeutic factor	Definition
Abreaction	Repressed material, particularly a painful experience, is brought back to consciousness
Acceptance	Feeling of being accepted by other members of the group
Altruism	Act of one member helping another; putting another's needs before one's own
Catharsis	Expression of ideas/thoughts/suppressed material that is emotional and produces relief
Cohesion	Sense that the group is working together toward a common goal
Consensual validation	Confirmation of reality by comparing own conceptualisations with other group members
Contagion	Expression of emotion by a member stimulates awareness of similar emotion in another
Corrective familial experience	Group recreates the family of origin to work through original conflicts psychologically
Empathy	Capacity of group members to put themselves in psychological frame of another member
Identification	Unconscious defense mechanism; characteristics of another person are incorporated
Imitation	Conscious emulation/modeling of one's behaviour after another; 'role modeling'
Insight	Conscious awareness/understanding of own psychodynamics and maladaptive behaviour
Inspiration	Process of imparting a sense of optimism to group members; 'instillation of hope'
Interaction	Free and open exchange of ideas and feelings among group members
Interpretation	Group leader formulates the meaning of a patient's resistance, defense, and symbols
Learning	Acquiring knowledge about new areas, such as social skills; receiving advice and obtaining guidance
Reality Testing	Ability of the person to evaluate objectively the world outside the self
Transference	Projection of feelings, thoughts, and wishes onto the therapist
Universalisation	Awareness of patients that they are not alone in having problems; 'self-disclosure'
Ventilation	Expression of suppressed feelings, ideas, or events to other group members
Adapted from Kaplan and Sadock's Synopsis of Psychiatry, 12 th Edition, 2022. p. 757.	

19. Definition about legally and morally responsible for action and inaction- culpability

- A. Mens Rea..
- B. Offense trailed by jury_ indictable
- C. Actus rea
- D. Culpability,
- E. Diminished responsibility...
- F. Criminal responsibility
- G. Indictable offence

Automatism one stem

[SPMM] You are asked to conclude whether automatism could be used as a defence. Which of the following statements is true?

Select one:

- A. Sane automatism cannot lead to total acquittal
- B. Sane automatism are due to internal causes
- C. In proven insane automatism, the accused cannot be found guilty by reason of insanity
- D. Insane automatism include epilepsy**
- E. Automatism is a common plea

Automatism: It is a rare plea. Legal classification includes sane automatism where the act is one off, often acquittal takes place. **Hypoglycemia (isolated), night terror and dissociative states can be classed as sane automatism.** Here the cause is often 'external'. Insane automatism are likely to recur ('internal') and the counsel can plead not guilty by reason of insanity (insanity defence). **Night walking, epilepsy, hypoglycemia due to a recurring condition are classed under this.** There is no medical sense in these classifications. Automatism is a plea by a criminally accused that his or her actions were not under the control of the conscious mind. In other words, claiming that one's bodily movements were not willfully executed and were involuntary.

Definitions in Forensic Psychiatry
Match each forensic term with its definition.

1. Actus reus		×
2. Mens rea		×
3. Culpable homicide		×
4. Involuntary		×
5. Voluntary		×
6. Indictable offence		×

Crime committed with intent
 Crime committed without intent
 Manslaughter
 Killing of an infant
 The (voluntary) act of crime
 The (evil) intent of crime
 Crime of dishonesty
 A serious offence tried in Crown court

EXPLANATION: According to criminal law, committing a socially harmful act is not the sole criterion of whether a crime has been committed. Instead, the objectionable act must have two components: actus reus and mens rea. Actus reus refers to the (voluntary) act of crime. Mens rea refers to the (evil) intent of crime. An evil intent cannot exist when an offender's mental status is so deficient, so abnormal, or so diseased to have deprived the offender of the capacity for rational intent. Neither behaviour, however harmful, nor the intent to harm, alone, is grounds for criminal action.

Culpable homicide (manslaughter) may be voluntary or involuntary. When voluntary, killing is done with the intent to murder, but partial defence applies (e.g. suicide pact, severe provocation). When involuntary, the conduct was either grossly negligent (given the risk of death), or the conduct was an unlawful act involving a danger or harm that resulted in death. An indictable offence is a serious offence tried in Crown court.

20. Akathisia treatment

- A. Propranolol**
- B. Tetrabenazine
- C. Reduce drug

	Dystonia	Pseudo-parkinsonism (e.g. tremor)	Akathisia	Tardive dyskinesia
Prevalence (with older drugs)	Approximately 10% But more common in - young males - neuroleptic-naïve - high potency drugs (e.g. haloperidol)	Approximately 20% But more common in - elderly females - those with pre-existing neuro damage (e.g. stroke)	Approximately 25%	5% of patients per year of antipsychotic exposure But more common in - elderly women - those with affective illness - those who have had EPSE early on in treatment
Time taken to develop	Acute dystonia can develop within minutes or hours of starting antipsychotics	Days to weeks after antipsychotic started or dose increased	Hours to weeks	Months to years
Treatment	• Anticholinergic drugs* • Switching to alternative antipsychotic • Botulinum toxin	• Reduce the dose • Anticholinergic drugs* • Switching to alternative antipsychotic	• Reduce the dose • Switching to alternative antipsychotic (quetiapine/olanzapine) • Propranolol • Mirtazapine or mianserin (low dose) • Anticholinergic drugs* • Cyproheptadine • Benzodiazepine (e.g. diazepam or clonazepam) • Clonidine	• Stop anticholinergic (if prescribed) • Reduce the dose • Switch to an atypical (clozapine and quetiapine have best evidence) • Tetrabenazine • Ginkgo biloba

21. Clozapine weight gain (no lifestyle medication as an option)

A. Metformin

B. Gliclazide

[SPMM] A male patient on clozapine has gained 20kg in 3 months, though his psychotic symptoms are well controlled. The next step is to

Select one:

A. Reduce the dose of clozapine

B. Initiate lifestyle modifications

C. Switch to high dose haloperidol

D. Switch to risperidone

E. Switch to quetiapine

Explanation: The most effective weight reduction interventions in people with schizophrenia have been shown to be **individual lifestyle counselling** and exercise interventions. Lifestyle modification should be considered the first line intervention. Following lifestyle counselling and exercise interventions, the next **most effective weight reduction interventions have been shown to be psychoeducation, aripiprazole augmentation, topiramate, d-fenfluramine, and metformin.**

Metformin is now considered to be the drug of choice for the prevention and treatment of antipsychotic-induced weight gain, although GLP- 1 agonists may ultimately prove more effective and better tolerated.

Management of physical health in schizophrenia

Outcome	Most effective interventions	Effective interventions
Weight reduction	Individual lifestyle counselling Exercise interventions	Psychoeducation Aripiprazole augmentation Topiramate D-fenfluramine Metformin
Waist circumference reduction	Aripiprazole augmentation Topiramate	Dietary interventions
Reducing glucose levels	Switching from olanzapine to quetiapine or aripiprazole Metformin	Glucagon-like peptide-1 receptor agonists Dietary interventions Aripiprazole augmentation
Improvement of insulin resistance	Metformin Rosiglitazone	
Triglycerides and LDL	Topiramate	Lifestyle interventions
Total cholesterol and HDL	Metformin	Lifestyle interventions (for total cholesterol)

Adapted from Vancampfort D et al. 2019



Management of medication side effects

Choose the best medication to be added in the management of each of the following scenarios.

1. A patient taking clozapine gains substantial weight. ✗
2. A patient with schizophrenia and diabetes develops subjective restlessness in his legs. ✗
3. A patient receiving treatment for schizophrenia experiences drooling and tightening of the facial muscles. ✗
4. A patient with longstanding schizophrenia develops involuntary, repetitive movements of her tongue and lips. ✗

EXPLANATION: **Metformin** is now considered to be the drug of choice for the prevention and treatment of antipsychotic-induced weight gain, although GLP-1 agonists may ultimately prove more effective and better tolerated.

There is limited evidence on the benefit-risk balance for any pharmacological treatment for **akathisia**, even those most commonly used, such as switching to an antipsychotic medication with a lower liability for the condition, or adding a beta-adrenergic blocker, a 5-HT_{2A} antagonist, or anticholinergic agent. The first-choice medication to add is usually low-dose **propranolol**, provided there are not contraindications (e.g. asthma, bradycardia, hypotension, etc.)

There are no medications licenced for the treatment of antipsychotic (especially clozapine)-induced hypersalivation, and many of the relevant published studies have limitations that preclude any

confident treatment recommendations. The evidence, such as it is, tends to favour anti-muscarinic agents such as propantheline and diphenhydramine.

Tetrabenazine (reversible VMAT2 inhibitor) is the only licensed treatment for moderate to severe TD in the UK. Depression, drowsiness, parkinsonism, and akathisia may occur. CYP2D6 genotyping may be required when higher doses are used. Reserpine is also effective but is rarely, if ever, used. Reserpine produces irreversible and nonselective, inhibition of both VMAT1 and VMAT2, resulting in blood pressure reduction, notable issues with gastric motility, impotence, and risk of prolonged depression, though some early evidence indicated a positive effect on psychotic symptoms per se.

ANSWERS: 1. – Metformin 2. – Propranolol 3. – Propantheline 4. – Tetrabenazine

22.Prevention:

- A. Universal
- B. Selective
- C. Indicated
- D. Secondary
- E. Tertiary

1. Low income families at risk of developing mental illness
2. Prevention for preventing bullying at school-- universal
3. At risk mental state—what sort of prevention

[SPMM] **Prevention in mental health**

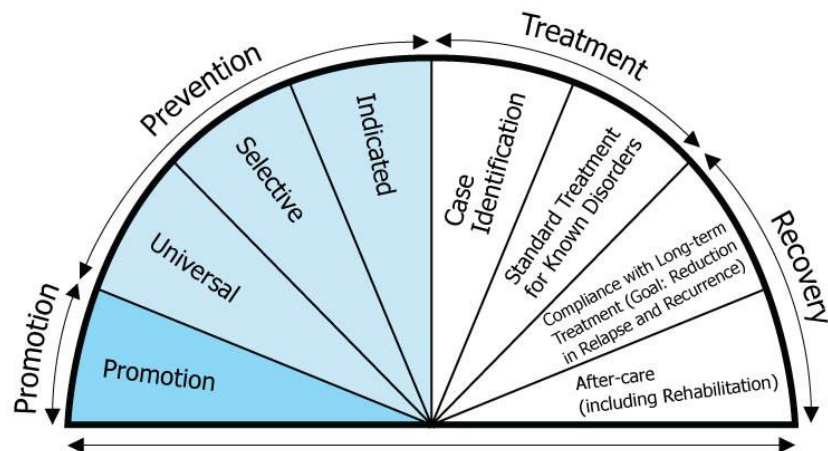
Choose the most appropriate term(s) from the options for each prevention scenario given below. Each option can be used more than once or not at all.

1. For individuals identified as having an "**at-risk mental state**" for psychosis, a prevention strategy is implemented offering Cognitive Behavioural Therapy (CBT) (choose TWO)
2. A school program aimed at providing educational and support services to low-income families to reduce the risk factors associated with mental health problems in children (choose TWO)
3. A one-day school-wide retreat program is designed to prevent bullying among students (choose TWO)

Indicated prevention
Primary prevention
Secondary prevention
Selective prevention
Tertiary prevention
Universal prevention
Promotion

CORRECT ANSWERS:

1. For individuals identified as having an "at-risk mental state" for psychosis prevention strategy is being implemented via offering Cognitive Behavioural Therapy (CBT) (choose TWO) - **Primary prevention; Indicated prevention**
2. A school program aimed at providing educational and support services to low-income families to reduce the risk factors associated with mental health problems in children. (choose TWO) - **Primary prevention; Selective prevention**
3. A one-day schoolwide retreat program is designed to prevent bullying among students. (choose TWO) - **Primary prevention; Universal prevention**



Prevention in mental health

Choose the most appropriate term(s) for each prevention scenario. Each option can be used more than once or not at all.

1. A boy identified as having an "at-risk mental state" for psychosis is offered Cognitive Behavioural Therapy (CBT). (choose TWO)

2. Patients with bipolar I disorder who have attempted suicide are offered lithium with the aim of preventing further suicide attempts.

(choose TWO)

3. A one-day schoolwide retreat program is designed to prevent bullying among students. (choose TWO)

4. A suicide helpline is established and advertised to the general public. (choose TWO)

Prevention strategies in mental health are categorized into various levels, including **primary** (before the onset of an illness), **secondary** (as soon as the illness begins), and **tertiary** (targeting those with established conditions).

Primary prevention strategies target a population or community to prevent the initial occurrence of a health condition. These can be **selective**, as in a school program for low-income families that aims to reduce risk factors before any specific mental health problems develop, or **universal** as in a program aimed at reducing bullying behaviour among all students irrespective of age, gender, or prior reports of bullying behaviour. **Indicated primary prevention** strategies target individuals who are at heightened risk of developing a specific condition. Individuals with an "at-risk mental state" for psychosis fall into this category, and providing CBT at this stage can help prevent the onset of full-blown psychosis. Promotion refers to strategies that create supportive environments and conditions to improve overall health and are not focussed on any specific problem group or behaviour. As we move from promotion to tertiary preventions (e.g., rehabilitation), the number of direct beneficiaries will reduce but the specificity of the interventions will increase.

CORRECT ANSWERS: 1. – Indicated, Primary. 2. – Selective, Tertiary. 3. – Primary, Universal. 4. – Secondary, Universal.

23.Imaging

1. Syphilis → cerebral gumma
2. FTD → atrophy in frontal lobe, hypoperfusion in frontal lobe in MRI
3. AZ → Medial temporal lobe atrophy
4. Parkinson dementia
5. Lewy Body

DAT Scan reduced dopamine in basal ganglia

SPECT

Findings on neuroimaging studies	
For each of the following diseases, choose the correct finding.	CT/MRI show ventricular dilation, particularly of the temporal horns, and thinning of the cortex CT/MRI show relative sparing of medial temporal lobes; frequent periventricular lucencies on MRI CT/MRI show atrophy of the basal ganglia, with 'boxing' of the caudate and dilatation of the ventricles SPECT shows temporal and posterior parietal hypoperfusion PET shows asymmetrically decreased dopamine activity in the basal ganglia
1. Normal pressure hydrocephalus	✗
2. Alzheimer disease	✗
3. Dementia with Lewy bodies	✗
4. Huntington disease	✗
5. Parkinson disease	✗

EXPLANATION:

In normal pressure hydrocephalus, CT/MRI show ventricular dilation, particularly of the temporal horns, and thinning of the cortex.

In Alzheimer disease, CT shows cortical atrophy, especially over parietal and temporal lobes, and ventricular enlargement. MRI shows

atrophy of the grey matter. SPECT shows temporal and posterior parietal hypoperfusion.

In dementia with Lewy bodies, CT/MRI show relative sparing of medial temporal lobes in most cases. Moderate increases in deep white matter lesions and frequent periventricular lucencies are seen on MRI.

In Huntington disease, CT/MRI show atrophy of the basal ganglia, with 'boxing' of the caudate and dilatation of the ventricles. PET shows

decreased metabolism in the basal ganglia.

In Parkinson disease (PD), PET using radiolabeled fluorodeoxyglucose or fluorodopa usually shows asymmetrically decreased dopamine

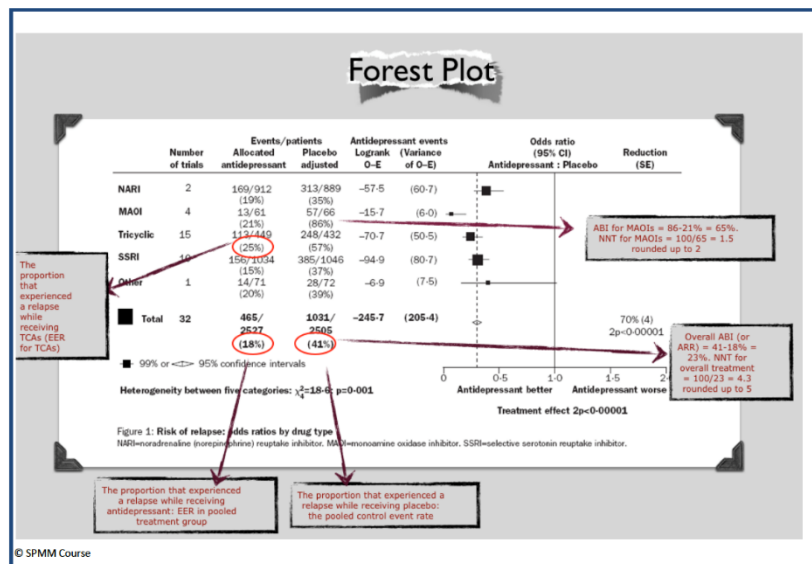
activity in the basal ganglia in pre-symptomatic individuals and in those with overt PD.

CORRECT ANSWERS:

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1. **Normal pressure hydrocephalus** – CT/MRI show ventricular dilation, particularly of the temporal horns, and thinning of the cortex
2. **Alzheimer disease** – SPECT shows temporal and posterior parietal hypoperfusion
3. **Dementia with Lewy bodies** – CT/MRI show relative sparing of medial temporal lobes; frequent periventricular lucencies on MRI
4. **Huntington disease** – CT/MRI show atrophy of the basal ganglia, with ‘boxing’ of the caudate and dilatation of the ventricles
5. **Parkinson disease** – PET shows asymmetrically decreased dopamine activity in the basal ganglia

24. Calculate NNT for SSRIs = $1/ARR$



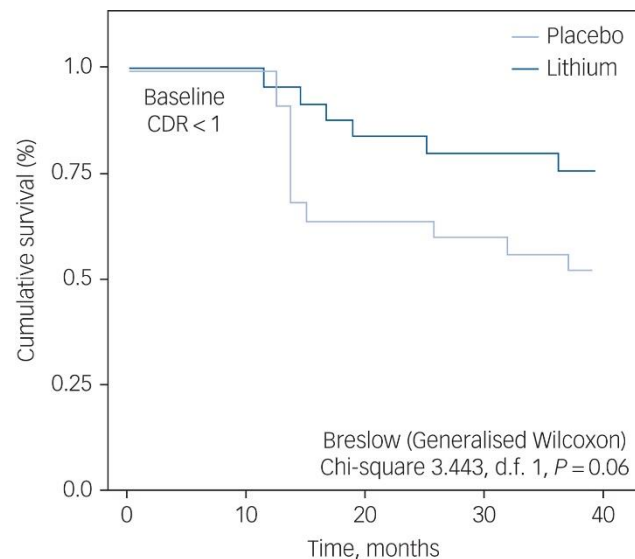
25. Question about MAOI

26. lithium at 10 months and 40 months

See Fresh question 2023 Q.15

27. Difference between placebo and lithium, 3 question from the above graph

See Fresh question 2023 Q.15



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28. Highest efficacy? **Amitriptyline**
See Mock 9 Q. 4

29. Which one had largest effect?? **Amitriptyline**

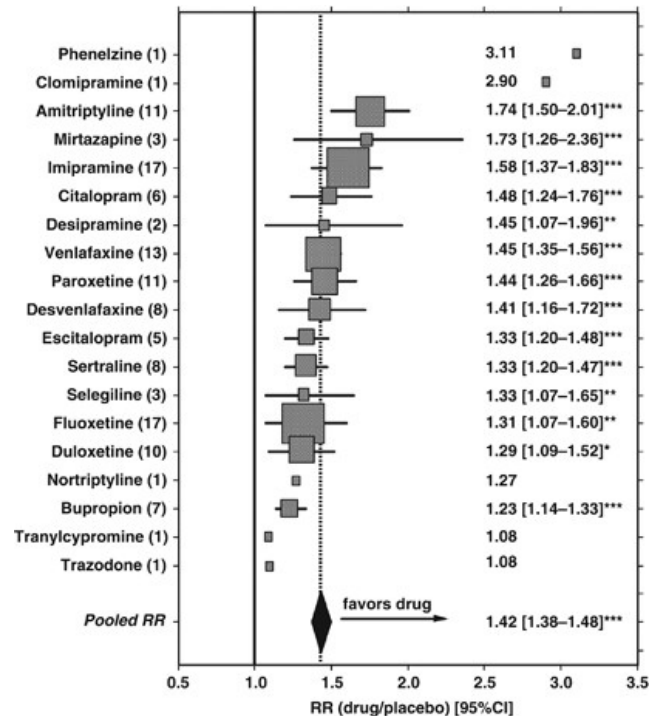
30. Which has more dispersion (*longest horizontal line/ widest CI*) ?? **Mirtazapine**

N.B: Largest weightage → **Imipramine and Fluoxetine** (square size (17))

1. Choose the antidepressant drug with **best efficacy when compared to placebo**. (Choose ONE)
2. Choose the antidepressant drugs with **highest weighting in the meta-analysis** (Choose TWO)
3. Which drug with $n > 2$ trials has the **smallest effect size**? (Choose ONE)

4. Choose an **antidepressant drug with trial results favouring placebo** (Choose ONE)
5. Choose the antidepressant drugs with highest number of eligible trials (Choose TWO)

Correct Answer : 1. Amitriptyline 2. Fluoxetine, Imipramine 3. Bupropion 4. None of them 5. Fluoxetine, Imipramine



31.1. Baby came with low birth weight, poor feeding, microcephaly, heart problem? Most probable disorder?

DiGeorge syndrome

2. Noisy baby - **Cri Du chat**

3. Gaze avoidance - **Fragile X**

- A. Down syndrome
- B. Angelman
- C. Williams
- D. Fragile x
- E. Cri du chat
- F. Turner syndrome

Behavioural phenotypes in intellectual disability (2)

Choose one diagnosis for each of the behavioural phenotypes.

Shyness, gaze avoidance, and social anxiety

Choose...

Compulsive, severe self-mutilating behaviours such as biting of the lips, inside of the mouth, and fingers

Choose...

Insatiable appetite, hyperphagia, and obesity

Choose...

Self-hugging posturing, self-injurious behaviours like head banging, aggression

Choose...

Hyperactivity, stubbornness, private speech

Choose...

Explanation: Behavioural features of **fragile X syndrome** appear to be similar to those seen in ADHD and autism, including hand flapping/waving, repetitive mannerisms, shyness, gaze avoidance, poor peer relationships, communication difficulties (e.g. delayed language development, conversational rigidity, perseveration, echolalia, palilalia, cluttering, and over-detailed/circumstantial speech), and psychiatric problems (e.g. prominent depression/anxiety).

In **Lesch-Nyhan syndrome**, severe compulsive self-mutilation is often seen, including biting of the lips, inside of the mouth, and fingers. The striking feature of Prader-Willi syndrome is massive hyperphagia with associated compulsive food-seeking, and consequent marked obesity. The behavioural phenotype of Prader-Willi syndrome also includes hoarding, impulsivity, emotional lability, tantrums, excessive daytime sleepiness, skin-picking, and aggression.

Behavioural problems in Smith-Magenis syndrome include 'self-hugging' posturing, aggression, self-injury, hyperactivity, severe sleep problems, and other autistic features.

The **behavioural phenotype of Down syndrome** includes being passive, affable, demonstrating hyperactivity in childhood, being stubborn, and often talking or mumbling to themselves (private speech).

Clinical syndromes & intellectual disability
Regarding the following, match the syndrome with the clinical situation:

1. A child suffered with disorder of purine metabolism. Select ONE syndrome ✖

2. A child with microcephaly and a high-pitched cry. Select ONE syndrome ✖

3. A child was born with tetralogy of Fallot. Select THREE syndromes ✖

✖ ✖

Correct Answer is : 1. Lesch-Nyhan syndrome 2. Cri du chat syndrome 3. Williams syndrome, Downs syndrome, DiGeorge syndrome

32.Calculation of sensitivity: **true positive / true positive + false negative**

33.Calculation of negative predictive value: **true negative / false negative + true negative**

34.Calculation of post-test odd, answer was not fitting in LR positive question

[SPMM] A screening instrument has a pre-test probability of 1 in 8 with 80% sensitivity and 80% specificity. Calculate the post-test odds of a positive test result.

Select one:

- A. 1 in 2
- B. 1 in 3
- C. 7 in 4
- D. 2 in 1
- E. 4 in 7**

EXPLANATION: To calculate the post-test odds, we need to first determine the pre-test odds and then apply the likelihood ratio.

STEP 1: To calculate Pre-test odds:

$$\text{Pre-test Odds} = \text{Pre-test Probability} / (1 - \text{Pre-test Probability})$$

$$\text{Pre-test Odds} = (1/8) / (1 - 1/8)$$

$$\text{Pre-test Odds} = (1/8) / (7/8)$$

$$\text{Pre-test Odds} = 1/7$$

STEP 2: To calculate the likelihood ratio (LR+):

$$\text{LR+} = \text{Sensitivity} / (1 - \text{Specificity})$$

$$\text{LR+} = 0.80 / (1 - 0.80)$$

$$\text{LR+} = 0.80 / 0.20$$

$$\text{LR+} = 4$$

STEP 3: Finally, to calculate the post-test odds:

$$\text{Post-test Odds} = \text{Pre-test Odds} \times \text{LR+}$$

$$\text{Post-test Odds} = (1/7) \times 4$$

$$\text{Post-test Odds} = 4/7$$

So, the correct post-test odds are **4 in 7**.

- **LR+ = sensitivity / 1- specificity**
- **LR- = 1- sensitivity / specificity**
- **1- specificity = false positive rate**
- **1- sensitivity = false negative rate**

35. Strongest correlation

- A. -0.9**
- B. 0.7
- C. 0.5

36. Correlation

37. Regression

38. Post-test odd calculation **See Above**

39. If we increase sensitivity of a test and cut-off is unchanged what will happen?

- A. Increase false positive
- B. Increase false negative

Cut-off point for positivity may be adjusted to optimize sensitivity and specificity for different purposes, which are inversely related (cut-off point with decreased sensitivity is associated with increased specificity and vice-versa)

[SPMM] If the sensitivity of a test increases which of the following occurs?

Select one:

- A. The cut-off point increases.
- B. The power of statistical test drops.
- C. The specificity increases.
- D. The false positive error rate increases.**
- E. The false negative error rate increases.

When sensitivity increases, more number of cases will be diagnosed despite not having the disease (an increase in false positive rate). When you select a lower criterion value, then the true positive fraction and sensitivity will increase. On the other hand, the false positive fraction will also increase, and, therefore, the true negative fraction and specificity will decrease.

ChatGPT If the sensitivity of a test increases while the cutoff point remains constant, it means that the test is better able to correctly identify true positive cases while maintaining the same threshold for classifying positive and negative results. Here's what happens:

1. Increase in Sensitivity: Sensitivity measures the proportion of true positive cases correctly identified by the test. So, an increase in sensitivity means that the test is more effective at detecting true positive cases among those who actually have the condition.
2. Constant Cutoff Point: The cutoff point is the threshold used to classify test results as positive or negative. It remains the same regardless of changes in sensitivity. This means that the criteria for determining whether a result is positive or negative remains unchanged.

Effects:

- Fewer False Negatives: With increased sensitivity, the test is better at correctly identifying individuals who have the condition (true positives), leading to fewer false negative results. This reduces the likelihood of missing cases that should be detected.
- Potential Increase in False Positives: While sensitivity improves, maintaining the same cutoff point may lead to an increase in false positive results. Some individuals who don't have the condition may now test positive because the test is more sensitive, but the cutoff point remains constant.
- Overall Impact: Depending on the context, the increase in sensitivity could be beneficial if correctly identifying true positive cases is crucial. However, it's essential to consider the trade-off with an increased rate of false positives, which could lead to unnecessary follow-up tests or treatments.

40. Increase prevalence of which disease in age 11-12?

- A. OCD**
- B. Tourette
- C. Depression

[SPMM] Which of the following disorders increases in severity and prevalence between ages 10-12 years-old?

Select one:

- A. Obsessive-compulsive disorder**
- B. Autism spectrum disorder
- C. Attention-deficit hyperactivity disorder

- D. Conduct disorder
- E. Tourette syndrome

There are two peaks of incidence of OCD, one early peak with a mean age of 10 (with a standard deviation of 2.5 years) years of age, so that two-thirds of affected youth will have an onset between about 7 and 12 years of age and well before adulthood. There is a second peak of incidence in the early 20's. OCD in children may go unrecognized for some time and there is often a lag between age at ascertainment and age at onset. Symptoms are often secretive and hidden due to embarrassment or shame. Paediatric OCD has relatively high rates of remission; approximately one third to one half remit or improve to subthreshold levels over time.

41. how many times likely ASD more common in prison population?

- A. 2 -3 times
- B. no change
- C. 4-5 times
- D. 8-9 times

Answer from recall file: in prison in UK 5-7 times

42. In BALANCE study they compared which drug against what?

[SPMM] With respect to prophylaxis in bipolar disorder, the BALANCE study showed which of the following findings?

Select one:

- A. Lithium is superior to valproate**
- B. Valproate is superior to lithium
- C. Lamotrigine is superior to lithium
- D. The combination of lithium and valproate is superior to lithium
- E. The combination of lithium and lamotrigine is superior to lithium

EXPLANATION: The independent BALANCE study found that **valproate as monotherapy was relatively less effective than lithium or the combination of lithium and valproate**, casting doubt on valproate's use as a single first-line prophylactic treatment in bipolar I disorder.

BALANCE found **lithium to be numerically superior to valproate**, and the combination of lithium and valproate to be statistically superior to valproate alone. Given this and the fact that valproate is not licensed for prophylaxis, it should be considered a second-line treatment.

43. what was the aim of CUTLASS 1 study? **1st generation and 2nd generation equal**

[SPMM] With respect to first-generation antipsychotics (FGAs) compared to second-generation antipsychotics (SGAs), which of the following was a major finding of the CUTLASS study?

Select one:

- A. SGAs are superior in efficacy to FGAs
- B. FGAs show greater improvement on quality of life
- C. FGAs are superior in efficacy to SGAs
- D. Efficacy is similar between FGAs and SGAs**
- E. SGAs show greater improvement on quality of life

EXPLANATION: In CUtLASS (Cost Utility of the Latest Antipsychotic drugs in Schizophrenia Study), a 12-month open-label trial, 277 patients were randomised to receive either an FGA or an SGA. Efficacy was similar between the two groups, with only limited improvement of psychopathology and quality of life. The authors concluded that SGAs do not markedly differ from FGAs regarding compliance, quality of life, or effectiveness.

44. Minimum age to start fluoxetine?

A. 5

B. 6

C. 7

D. 8

☞ Sertraline at age 6

☞ Fluvoxamine at age 8

45. Stigma what type of cost?

Costs and economic studies

You are conducting an economic study comparing the cost-effectiveness of medication-based treatment to non-medication-based treatment (residential care in a therapeutic open community) for chronic PTSD. For each of the questions below, choose the appropriate description from the provided list.

1. You want to describe the additional cost incurred for the medication-based treatment over the non-medication-based treatment.

✖

2. A junior doctor is temporarily reassigned to the therapeutic open community unit that is providing the non-medication-based treatment. As this unit is at a geographically distant location, the hospital is paying for the commuting expenses of the doctor. Which cost concept best represents the financial impact of this process? ✖

3. Patients receiving the therapeutic open community care are required to be absent from work due the need for residential stay, resulting in a loss of productivity and wages. What type of cost is associated with the days lost to sickness? ✖

4. During the trial, patients who receive medications that result in excessive sleep report being stigmatised by their family members for being inactive. Some of them find this distressing and believe that this contributes to difficulties in making living arrangements. What type of cost is associated with this phenomenon? ✖

CORRECT ANSWERS: 1. Incremental cost - 2. Direct cost - 3. Indirect cost - 4. Intangible cost

Incremental cost refers to the additional cost incurred when comparing two alternative treatments or interventions. In this case, it quantifies the extra cost associated with using medication-based treatment compared to non-medication-based treatment.

Direct cost represents the specific, measurable expenses incurred because of a particular activity in the course of provision of care. In this scenario, the additional commuting expenses directly relate to the junior doctor's temporary reassignment, making it a direct cost for the health system.

Indirect cost refers to the costs that are not directly tied to a specific activity but arise because of that activity.

In this case, the days lost to **Intangible** cost is a 'conceptual' cost, not something that is readily quantifiable in euros or pounds but are still significant and impactful. It represents costs that are challenging to quantify in monetary terms because they involve non-material aspects, such as emotional distress or stigma. The economic impact of stigma associated with mental health conditions often falls into the category of intangible costs.

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The attached figure is from Fautrel B, Boonen A, de Wit M, Grimm S, Joore M, Guillemin F. Cost assessment of health interventions and Diseases sickness result in a loss of productivity and wages for the patient, which are indirect costs of the treatment, borne by the patients.

46. Cost of paying psychologist, what type of cost? **Direct**

47. Cost of productivity lost?

[SPMM] In an attempt to study the cost of adherence vs. non-adherence to treatment in depression treated at primary care centres, an investigator calculates the costs of illness in terms of the costs for hospitalization, GP-visits, specialist referral, cost of medications, and cost of each day of sick leave due to depression. Which of the following among the above can be called an indirect cost?

Select one:

- A. Cost of specialist referral
- B. Cost of medications
- C. Cost of hospitalisation
- D. Cost of GP visits
- E. Cost of sick absence**

Direct costs: Cost of hospital space, cost of drugs, equipments, salary of doctors and other professionals etc. **Indirect costs:** Costs due to **loss of productivity** for the patient, cost of having a caretaker for one's dog or a nanny for one's child when hospitalised etc. Intangible (not measurable) costs: Pain and suffering, social stigma, physical discomfort, etc.

48. Cost of psychotherapy minus cost of medication. What sort of cost? **Incremental cost**

49. Definition of automatism.

Automatism [Psych-Mentor]

Automatism is a defence used when the act is believed to have occurred unconsciously. It comes in two forms:

- **Sane automatism** - where the act occurs from an **external** cause such as a head injury.
- **Insane automatism** - where the act occurs from an **internal** cause such as epilepsy

[Psych-Mentor] Which of the following would qualify for a defence of insane automatism?

- A. Drink being spiked with LSD
- B. Somnambulism**
- C. Provocation
- D. Concussion
- E. Alcohol intoxication

[Psych-Mentor] Which of the following would qualify for a defence of sane automatism?

- A. Somnambulism
- B. Cerebral tumour
- C. Hyperglycaemia resulting from uncontrolled diabetes
- D. Concussion**
- E. Epilepsy

50. Doctor from one ward to another: **opportunity cost**

[SPMM] Due to the difficulty in recruiting a full-time nurse for the ECT clinic, your hospital decides to provide the nurse who works at pain clinic to attend the ECT service on 'secondment'. Select the term that best describes the costs incurred by patients due to the absence of a nurse at the pain clinic **as a result of this arrangement**.

Select one:

- A. Indirect cost
- B. Direct cost
- C. Zero-sum cost
- D. **Intangible cost**
- E. Opportunity cost

The opportunity cost is defined as the benefits that must be foregone when making any **resource allocation decision**. It refers to the fact that expending resources on one health care issue inevitably reduces the investment that could be made elsewhere. The opportunity cost of employing an ECT nurse is the benefit we forego by being unable to support the pain clinic. **If the question asks about the cost borne by the patient then it will be Intangible.**

51. Hypertelorism, hypotonia, moderate ID in which disease? Cri-du chat

[SPMM] A 5-year-old boy with severe intellectual disability has a round face with **hypertelorism**, slanting palpebral fissures, broad flat nose, and an infantile cat-like cry. Which of the following conditions is he most likely to suffer from?

Select one:

- A. **Cri-du-chat syndrome**
- B. Down syndrome
- C. Angelman syndrome
- D. Smith-Magenis syndrome
- E. Williams syndrome

Explanation: Cri-du-chat syndrome is a partial monosomy; karyotype 5p- (varies from deletion of a small band at 5p15.2 to the entire arm of 5p). The infant has a characteristic **high-pitched cry** that resembles a cat. Clinical features include **microcephaly**, **rounded face**, **hypertelorism**, **micrognathia**, dental malocclusion, epicanthic folds, low-set ears, **hypotonia**, and severe/profound intellectual disability.

Hypertelorism refers to an abnormally increased distance between two organs or bodily parts, usually referring to an increased distance between the orbits (eyes). It is seen in some conditions like Di George's syndrome, Noonan syndrome, Edwards syndrome, cri du-chat syndrome and neurofibromatosis. Salient features of DiGeorge syndrome can be summarized using the mnemonic CATCH-22, with the 22 reminding us of chromosome 22, and CATCH referring to Cardiac Abnormality, Abnormal facies (palatal abnormalities, hypertelorism), Thymic aplasia, Cleft palate, Hypocalcaemia / Hypoparathyroidism

52. Jocular speech, gaze aversion which disease?

Fragile X syndrome is characterized by social anxiety, including poor eye contact, **gaze aversion**, prolonged time to commence social interaction, and challenges forming friendships. Social anxiety is one of the most common features seen, with up to 75% of males in one series characterized as having excessive shyness and 50% having panic attacks. It is thought to be due to impairment of face encoding and recognition.

Behavioural features include hyperactivity, inattention, anxiety, stereotypes, **speech and language delays**, IQ decline, repetitive mannerisms, shyness, gaze avoidance, poor peer relationships, and communication difficulties. [SPMM]

53. How to turn qualitative to quantitative study? **log transformation,**

[SPMM] **Transformation of data:**

Parametric statistical tests offer more robust analysis than non-parametric tests. Hence, one may convert data into a form more suitable for parametric analysis using transformation methods. The most commonly used method is **log transformation**. For a right skew (positive) this yields a normal distribution (called lognormal curve). Such transformation can also linearize a nonlinear correlation with the upward shift. **Square root transformation** has normalizing and linearizing properties; in addition it can stabilize variance. It is used for normalizing Poisson distributions.

Other types include

- **Reciprocal transformation,**
- **Square transformation, and**
- **Logit transformation.** The **reciprocal transformation** is often used for analysis of survival rates. The logit transformation is often used for distribution of proportions. It linearizes a sigmoid curve. Note that confidence intervals may be difficult to interpret after transformation. Geometric mean values may get distorted in the process to some extent.

54. Scenario on nested case control study? What type of study it was? **Nested was answer for one**

Question 24 Mock 11

[SPMM] **Antipsychotic Drugs and Hyperglycemia in Older Patients With Diabetes Background:** Evidence suggests that there is an association between antipsychotic drugs and new-onset diabetes, but little is known about the risk of hyperglycaemia among persons with pre-existing diabetes. **Methods:** Using population-based health databases in Ontario, Canada, persons aged 66 years or older with diabetes who started treatment with an antipsychotic drug from April 1, 2002, to March 31, 2006, were followed up from treatment start until March 31, 2007. The cohort was subdivided into 3 groups: insulin-treated, an oral hypoglycaemic agent only -treated, and no diabetes treatment. We defined cases as patients hospitalized (emergency department visit or hospital admissions) for hyperglycaemia. Each case was matched with up to 10 controls for age, sex, year of cohort entry and duration of follow-up. The authors compared the likelihood of hyperglycaemia among current users of atypical and typical antipsychotic agents with that among remote antipsychotic users (discontinued >180 days), based on prescriptions closest to the event date.

Results: Of 13 817 patients studied, 1515 (11.0%) were hospitalized for hyperglycaemia. Current antipsychotic treatment was associated with a higher risk of hyperglycaemia compared with remote antipsychotic use in all diabetes treatment groups (overall adjusted rate ratio, 1.50; 95% confidence interval, 1.29-1.74). The risk was increased among patients who were treated with atypical and typical

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antipsychotic agents and was extremely high among patients who were just starting treatment (only 1 prescription before event).

Conclusions: Among older patients with diabetes, the initiation of treatment with antipsychotic drugs was associated with a significantly increased risk of hospitalization for hyperglycaemia ($P < .001$). The risk was particularly high during the initial course of treatment and was increased with the use of all antipsychotic agents. What type of study is this?

Select one:

- A. Prospective cohort study
- B. Case control study
- C. Retrospective cohort study
- D. Nested case control study**
- E. Cross-sectional survey

This study uses a cohort of individuals to separate them into cases and controls based on an 'outcome' event. This is a nested case-control study.

55. A young boy hitting his own head, most appropriate what is the next step...

- A. Start risperidone
- B. Do antecedent, behaviour analysis**
- C. Lorazepam
- D. Haloperidol

[SPMM] A person with intellectual disability has aggressive behavioural issues. Which of the following is the best first step in approaching this situation?

Select one:

- A. Functional behaviour assessment/analysis**
- B. Behaviour support plan
- C. Prescription of an antipsychotic
- D. Psychoeducation
- E. Physical restraints

Behaviour disorders are over-represented in intellectual disability (ID) populations. It is important to perform a thorough assessment, including functional assessment of the behaviour (i.e. description of the behaviour, situations, consequences, and reinforcers) before implementing management strategies. Originating in the mid-1960's, functional behavioural analysis has become the gold standard for understanding the environmental variables that come to shape and maintain problematic behaviours such as aggression, self-injury, and property destruction.

56. What percentage of schizophrenic patients commit homicide? **5 percent**

Forensic Psychiatry & Homicide

Choose the most likely data pertaining to epidemiological observations for the following statements:

The proportion of homicide perpetrators with a primary diagnosis of a personality disorder that received prison sentence

What percentage of homicide perpetrators with a primary diagnosis of a personality disorder had no contact with the mental health services in the year before?

What percentage of PD patients had no other psychiatric illness at the time of committing the homicide?

What percentage of homicide perpetrators with a history of 'mental disorder' has personality disorders as a primary diagnosis?

Choose...

- 45%
- 2-5%
- 9%
- 25-30%
- 50%
- 1-2%
- 5-10%
- 7%

}

Explanation:

All of the required data to answer this EMI can be found in "Avoidable Deaths" - five-year report of the national confidential inquiry into suicide and homicide by people with mental illness.

Among all homicide perpetrators, **30% have one or other mental disorder** mentioned in the psychiatric reports, with around 10% having had a contact with mental health services in the last 12 months. Interestingly schizophrenia, alcohol /substance use and affective disorders are equally prevalent (18% each, of all those 30% with a 'diagnosis'), but the most common diagnosis is personality disorders (27%). In essence, 5% of all homicide perpetrators have a diagnosis of schizophrenia and another 5% have personality disorders as the primary diagnosis. Of the 5% perpetrators with a personality disorder, 29% had been in contact with services in the last 12 months, 28% had been in contact but not in the last 12 months and 43% had no previous contact with services at all. In total, 55% of the perpetrators with a PD had a secondary diagnosis, mainly alcohol and drug dependence. 92% of those with PD as a primary diagnosis are imprisoned, while 4 % receive hospital.

57. Which one less likely cause discontinuation syndrome? **Fluoxetine**

Swapping antidepressants
For the clinical situations given below, choose the best option

1. A 45-year-old woman on paroxetine for 4 weeks has now decided to take citalopram, as she responded well to this medication in the past. (Choose ONE) ✖

2. A 52-year-old woman on Venlafaxine developed insomnia & dizzy spells. She is not keen to continue taking this antidepressant. She is now willing to take mirtazapine. (Choose ONE) ✖

3. A 57-year-old woman on fluoxetine has atypical depression. Now decided to treat her on moclobemide. (Choose TWO)
 ✖ ✖

4. A 46 year old man with depression on tranylcypromine has decided to switch over to sertraline (choose TWO)
 ✖ ✖

Withdraw and wait 1 week before starting new drug
 Withdraw and wait 2 weeks before starting new drug
 Withdraw and wait 5 weeks before starting new drug
 Withdraw and then start new drug straightaway
 Stop drug immediately
 Cross taper with caution
 Do not co-administer
 It may cause withdrawal symptoms
 Discontinue over a month

Correct Answer is : 1. Cross taper with caution 2. Cross taper with caution 3. Withdraw and wait 5 weeks before starting new drug, Do not co-administer 4. Withdraw and wait 2 weeks before starting new drug, Do not co-administer

When swapping from one antidepressant to another, abrupt withdrawal should usually be avoided. Cross tapering is preferred, where the dose of the ineffective or poorly tolerated drug is slowly reduced while the new drug is slowly introduced. When you switch from fluoxetine to moclobemide, withdraw and wait at least 5 weeks, as fluoxetine has the longest half-life of all SSRIs. When you switch from MAOI like tranylcypromine to SSRI, withdraw and wait for 2 weeks. MAOIs should not be co-administered with SSRIs, as this combination could precipitate serotonin syndrome.

[SPMM] The most characteristic symptom of SSRI discontinuation syndrome is
 Select one:

- A. Tremors
- B. Visual hallucinations
- C. Confusion
- D. Dizziness**
- E. Mania

The abrupt withdrawal of SSRI especially paroxetine or fluvoxamine (shorter half-life), is associated with dizziness, nausea, headache, depression, anxiety, insomnia, poor concentration, flu-like symptoms, paresthesias, and migraine-like symptoms. It usually requires at least 6 weeks of treatment before discontinuation and resolves spontaneously in 3 weeks. Those who have significant SSRI intolerance during treatment onset will have more discontinuation reactions. **Fluoxetine is the SSRI least likely to cause withdrawal syndrome as its metabolite has t_{1/2} more than 1 week leading to slow self-tapering in plasma.** Fluoxetine in some cases can treat discontinuation syndrome or used to prevent it when stopping an SSRI suddenly. But delayed withdrawal syndrome can occur with fluoxetine too.

58. More accurate way of reporting of sensitivity and specificity of study ?

A. STARD.

B. SRSQ.

C. PRISMA.

D. QUOROM.

E. CONSORT.

Guideline	Topic of guidelines
CONSORT (Consolidated Standards of Reporting Trials)	Randomised controlled trials
TREND (Transparent Reporting of Evaluations with Non-randomized Designs)	Non-randomised controlled trials
PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses)	Systematic Reviews and Meta-Analyses (Replaced QUORUM - Quality of Reporting of Meta-Analyses)
MOOSE (Meta-analysis Of Observational Studies in Epidemiology)	Observational Studies in Epidemiology
STROBE (Strengthening the Reporting of Observational Studies in Epidemiology)	Observational Studies in Epidemiology
SQUIRE (Standards for Quality Improvement Reporting Excellence)	Quality improvement studies
STARD (Standards for the Reporting of Diagnostic accuracy studies)	Diagnostic studies
MIAME (Minimum Information about a Microarray Experiment)	Microarray studies
COREQ (Consolidated criteria for reporting qualitative research)	Qualitative studies

Acamprosate = Craving
Disulfiram = Deterrent

59. Pt had alcohol 48 hours ago, where is disulfiram contraindicated? Heart failure,

[SPMM] The drug licensed to prevent relapse to alcohol use and has been found to have a modest treatment effect is

Select one:

- A. Naltrexone
- B. Disulfiram
- C. Acamprosate**
- D. Diazepam
- E. Chlordiazepoxide

Acamprosate is the drug licensed to prevent relapse to alcohol use and has been found to have a modest treatment effect. It is best suited to supporting abstinence in individuals who are concerned that craving will lead to a lapse/relapse. It should be commenced as soon as possible after abstinence has been achieved and should be maintained if the patient relapses. Note that **disulfiram is not a relapse reduction agent, but a deterrent**.

[SPMM] Mr Gary Cleaver is a 42-year-old man who has just completed a opioid detoxification and has remained abstinent from all opiates for the last 2 weeks. His wife is very supportive and his social situation is relatively stable. Which of the following would be most appropriate for him now?

Select one:

- A. Acamprosate
- B. Buprenorphine
- C. Inpatient rehabilitation
- D. Community rehabilitation
- E. Naltrexone**

Naltrexone is an opioid receptor antagonist that is indicated for use as an adjunct to aid formerly opiate dependent patients who remain abstinent now. It blocks the euphoric effects of opiates but patients must also have been abstinent for 7-10 days prior to prevent it precipitating severe withdrawals.

Contraindications include - psychosis, suicidal risk, severe personality disorder, consumption of alcohol, previous history of CVA, and uncompensated cardiac failure.

[SPMM] Which of the following is true about relapse prevention in alcohol dependence?

Select one:

- A. Acamprosate is not licensed for the elderly**
- B. If a patient relapses on acamprosate it must be stopped at once
- C. Smoking is contraindicated when taking disulfiram
- D. Acamprosate and disulfiram must not be combined
- E. Naltrexone has no place in treating alcohol dependence

Explanation: Safety and efficacy of acamprosate have not been established in patients younger than 18 years or older than 65 years. Acamprosate should not be administered to children, adolescents, or the elderly.

60. Which of the following is likely to be irreversible condition?

- A. Akathisia
- B. Eps
- C. Tardive dyskinesia**

61. Type of memory loss in ECT: **Explicit**

[SPMM] What type of memory is most affected by ECT?

Select one:

- A. Implicit memory
- B. Retrograde memory**
- C. Semantic memory
- D. Episodic Memory
- E. Anterograde memory

Retrograde memory (**Explicit memory**) i.e. the recall of events that occurred before the treatment, is the most commonly affected aspect of memory following Electro-Convulsive Therapy (ECT). It is an early side effect, occurs in **64%** of patients receiving ECT and usually resolves completely.

62. which is the non-reversible dementia?

- A. Wilson
- B. MND this is the answer**
- C. Pellagra
- D. Chronic subdural hematoma

[SPMM] Which of the following is an irreversible cause of dementia?

Select one:

- A. Chronic subdural haematoma
- B. Motor neuron disease**
- C. Normal pressure hydrocephalus
- D. Wilson disease
- E. Pellagra

EXPLANATION: Normal pressure hydrocephalus, chronic subdural haematoma, and pellagra (niacin deficiency) are all reversible causes of cognitive impairment/dementia. Early diagnosis and treatment can also reverse the manifestations of Wilson disease, including cognitive impairment/dementia. No current treatment can cure or even arrest motor neuron disease/amyotrophic lateral sclerosis. About 10-20% of patients with motor neuron disease meet the criteria for dementia. This group generally displays some clinical and pathological features of frontotemporal dementia.

63. Something with auditory processing disorder, boy perform bad in school due to noisy condition.. But in home does will correctly

[SPMM] Simon Lovell is a 10-year-old boy who struggles to concentrate at school and becomes upset in loud environments. However, he is happy at home and can produce excellent work at home without difficulty. Which of the following is the most likely diagnosis?

Select one:

- A. Auditory processing disorder**
- B. Dyslexia
- C. Social communication disorder
- D. ADHD
- E. ASD

Auditory processing disorder is a disorder that affects the way the brain processes auditory information. Individuals usually have normal structure and function of the outer, middle, and inner ear. However, they cannot process the information they hear in the same way as others do, which leads to difficulties in recognizing and interpreting sounds, especially the sounds composing speech.

64. Something with reactive attachment disorder - 18 months girl

[SPMM] Kate Forlock is a 2-year-old girl who was adopted into her current family around 9 months ago. Her adoptive mother has raised concerns that Kate does not seem to respond to affection, does not seek out cuddles and seems to avoid eye contact at times. She plays imaginatively and her language skills are developing relatively normally. Which of the following is the most likely diagnosis? Select one:

- A. **Reactive attachment disorder**
- B. ASD
- C. Auditory processing disorder
- D. Social communication disorder
- E. ADHD

Reactive attachment disorder is characterized by markedly disturbed and developmentally inappropriate ways of relating socially in most contexts. It can take the form of a persistent failure to initiate or respond to most social interactions in a developmentally appropriate way. It arises from a failure to form normal attachments to primary caregivers in early childhood. Such a failure could result from severe early experiences of neglect, abuse, abrupt separation from caregivers between the ages of six months and three years, frequent change of caregivers, or a lack of caregiver responsiveness to a child's communicative efforts.

[SPMM] A 5-year-old child who has been placed in multiple foster care homes since the age of 2 now shows little regard to seek approval from an adult caregiver before wandering away and a notable willingness to go off with unfamiliar adults with little hesitation. In ICD-11, these are core features of Select one:

- A. Conduct disorder
- B. Post-traumatic stress disorder
- C. **Disinhibited social engagement disorder**
- D. ADHD
- E. Reactive attachment disorder

Explanation: Disinhibited social engagement disorder (DSED) is newly described in both DSM-5 and ICD-11. It is characterised by grossly abnormal social behaviour, occurring in the context of a history of grossly inadequate child care (e.g., severe neglect, institutional deprivation). These children typically display reduced/absent reticence in approaching/interacting with unfamiliar adults; overly familiar verbal or physical behaviour; diminished/absent checking back with an adult caregiver; and willingness to go off with unfamiliar adults with little to no hesitation. **In contrast to reactive attachment disorder, symptoms of DSED tend to be more persistent following the provision of appropriate care.** Children with DSED related to repetitive maltreatment (e.g., chronic physical or sexual abuse) are at risk for developing co-occurring PTSD or complex PTSD. General impulsivity is commonly associated with DSED, and there is a high rate of co-occurrence with ADHD.

65. Heritability of ADHD - 70 to 80 percent.

[SPMM] Which of the following childhood psychiatric conditions has the highest heritability factor?

Select one:

- A. Conduct disorder
- B. ADHD**
- C. Separation anxiety disorder
- D. Oppositional defiant disorder
- E. School refusal

Explanation: ADHD has the highest heritability among the listed conditions. ADHD has a heritability of **70-80%**, and the risk of ADHD in siblings is 2-3x increased. Rates are increased in premature babies (especially extreme prematurity), babies born to mothers who used substances during pregnancy (e.g. tobacco or alcohol), following head injury, and in some genetic and metabolic disorders. None of the proposed prenatal risk factors have been confirmed as causal for ADHD.

66. Bournewood case, Liberty

[SPMM] What is the significance of the Bournewood case?

Select one:

- A. Deprivation of liberty**
- B. Medical negligence and safeguarding doctors
- C. Duty to warn
- D. Doctrine of necessity
- E. Treatment of minors without parental consent

The **Bournewood** judgement concerned a man with autism and intellectual disability who lacked capacity to consent to admission to hospital. As a result, amendments to the Mental Capacity Act were passed to introduce Deprivation of Liberty Safeguards (DoLS). DoLS require that, where an individual who lacks capacity is deprived of their liberty, an application should be made to a supervisory body for authorization to accommodate the individual. The supervisory body assesses the person's capacity and best interests, and reviews the planned care, and determines where detention is appropriate. The Bournewood ruling provides extra protection for the human rights of people who lack capacity and find themselves deprived of their liberty. This includes people with intellectual disabilities, mental health conditions, and older people in care homes.

The duty to warn the 3rd parties in case of imminent danger is illustrated by: [Psych-Mentor]

Select one:

- A. The Caldicott guardian
- B. Tarasoff's case**
- C. The Bamford review
- D. Bournewood case
- E. Genovese effect

Physicians and therapists in many jurisdictions have a duty to warn third parties of threatened danger arising from a patient's violent intentions. If the risk to third parties is so serious that it outweighs the

patient's privacy interest then the appropriate person or authority should be informed without undue delay. (Tarasoff case)

Q. Which of the following cases led to the development of the Fraser Guidelines?

- A. Reid
- B. Bournemouth
- C. Cowart
- D. Gillick**
- E. Stone

Correct answer : **Gillick**

Q. Which of the following cases highlights the role of the police in informing potential victims?

- A. Walton
- B. Bournemouth
- C. Pritchard
- D. Osman**
- E. Church

Correct answer : **Osman**

Osman warnings: The case of Osman vs UK concerned a teacher who was obsessed with one of his pupils (Osman). The obsession resulted in the death of Osman's father and in Osman being wounded. In this case the police were criticised to have not warned the family when they had been in possession of information that could have informed the family about the danger that Osman's teacher presented to them

67. A question on a girl fought with partner and who overdosed on three drugs(ssri, antipsychotic n i think ? Procyclidine) n had agitation, dry mouth, red eyes, What is the diagnosis ? **Delirium.**

Following a fight with her boyfriend, a 19-year-old woman overdoses on an over-the-counter drug. She presents with a flushed face and dry skin, and she appears irritable and agitated. Her MMSE score is 21/30. What is the most likely presenting diagnosis?

Select one:

- A. Delirium**
- B. Acute stress reaction
- C. Adjustment disorder
- D. Acute and transient psychotic disorder
- E. Borderline personality disorder

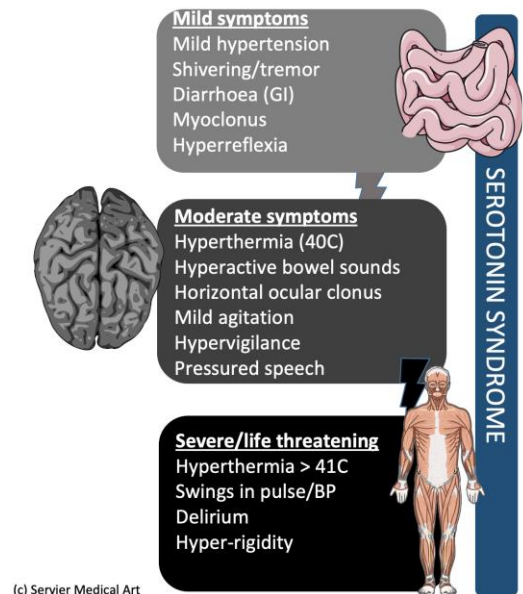
Delirium is characterised by a disturbance of attention, orientation, and awareness that develops within a short period of time, typically presenting as significant confusion or global neurocognitive impairment, with transient symptoms that may fluctuate depending on the underlying causal condition or aetiology. Delirium often includes disturbance of behaviour and emotion, and may include impairment in multiple cognitive domains.

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[SPMM] Mr Oliver Hirsh is a 24-year-old man who has a diagnosis of paranoid schizophrenia. He was brought to A&E by his family after he presented as confused at home. In A&E he was found to be hyperthermic, tachycardic, hypertensive and showing signs of increased muscular reflexes. He managed to inform the staff that he took an overdose of Sertraline and Haloperidol earlier that day. Which of the following is the most likely diagnosis? Select one:

- A. Serotonin syndrome**
- B. Anticholinergic toxicity
- C. Catatonia
- D. Sepsis
- E. Neuroleptic malignant syndrome

This is more likely to be Serotonin syndrome as the patient is presenting with rapid onset symptoms and hyperreflexia. Neuroleptic malignant syndrome is associated with onset of symptoms over the course of days to weeks and characterized by muscular rigidity. Anticholinergic toxicity is associated with profound visual impairment, urinary retention and marked constipation.



[SPMM] Which one among the following is a significant risk factor for the development of Neuroleptic malignant syndrome? Select one:

- A. Female Patient
- B. Low ambient temperature
- C. Being on oral medication
- D. Dehydration**
- E. History of personality disorder

Risk factors for NMS: High ambient temperature, dehydration, patient agitation or catatonia, rapid antipsychotic initiation/dose escalation, **withdrawal of anti-parkinsonian medication**, use of high potency agents/depot IM preparations, history of organic brain disease such as dementia, history of affective disorder, previous history of NMS, Predisposing drugs like lithium and anticholinergic medications.

68.Crimes of passion question having amnesia

[SPMM] Amnesia after a violent offence: Select one:

- A. Is associated with domestic homicide ('crimes of passion')**
- B. Is often seen in malingering
- C. Renders a person unfit to plead
- D. Is associated with impaired neuropsychological functioning
- E. Often resolves after a trial is over

Amnesia for a violent offense is associated with domestic homicide (killing a spouse, lover, or partner) and dissociative symptoms at the time. It is not associated with impaired neuropsychological

functioning. Amnesia of an alleged offense is irrelevant to fitness to plead, but short-term memory failure due to organic impairment may be such as to make following proceedings in court impossible.

[SPMM] Homicide-suicide is a rare but serious event when an individual will kill another person, usually current or past partners, before ending their own life. Which of the following mental disorders is most associated with this event?
Select one:

- A. Substance misuse
- B. Depression**
- C. Personality disorder
- D. Anxiety disorder
- E. Schizophrenia

The most common mental health diagnosis associated with homicide-suicide in intimate or family relations, is depression. Other diagnoses are rare. Case series report that the typical homicide-suicide case involves a depressed, controlling husband who shoots his ill wife.

69. Pt took antipsychotic, antidepressants, and other drugs , Malignant hyperthermia, NMS, serotonin syndrome - related stem, hyperthermia, rigidity—had myoclonic jerks ,ankle clonus
Formulas

70.CER-EER over CER is what?

Options were

- A. NNT
- B. NNH
- C. ABI
- D. RR
- E. RBI**

Event Rates in Clinical Trials

Lead In: Which of the following terms best describe the concepts below?

1. Experimental Event Rate- Control Event Rate ✖
2. $1 / (\text{Experimental Event Rate} - \text{Control Event Rate})$ ✖
3. Proportion of experimental event/proportion of control event ✖

Explanation: 1. The absolute arithmetic difference in rates of observed outcomes of interest between experimental and control participants in a trial, calculated as $|\text{EER} - \text{CER}|$ is the risk difference. It is also called the ARR or ABI (Absolute Benefit Increase)= $\text{CER} (\text{Control Event Rate}) - \text{EER} (\text{Experimental Event Rate})$.

2. The Number Needed to Treat (NNT) is the number of patients you need to treat to prevent one additional bad outcome (death, stroke, etc.). To calculate the NNT, you need to know the Absolute Risk Reduction (ARR); the NNT is the inverse of the ARR: $\text{NNT} = 1/\text{ARR}$, NNTs are always rounded up to the nearest whole number.

3. The ratio of risk in the treated group (EER) to the risk in the control group (CER) is called Relative Risk or Risk Ratio (RR). This is often used in RCTs and cohort studies and is calculated as EER/CER .

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71. Fisher exact when to use?? **Answer less than 5**

Statistical tests I

For each of the following studies, choose the most appropriate statistical tests:

1. 22 patients with chronic post-traumatic stress disorder (PTSD) were treated with naltrexone 100-200 mg/day and mean score of hyper arousal and avoidance symptoms were compared before and after 6 weeks of treatment. (Choose one)

2. In Helsinki High Risk study, the proportion of high-risk offspring (born to mothers with schizophrenia) who develop neurological soft signs was compared with the proportion of low risk offspring (control mothers) who develop them. (Choose two)

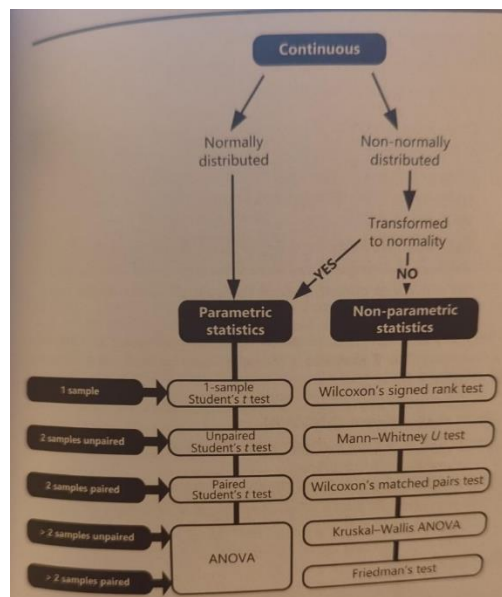
3. Serum epidermal growth factor levels were compared between subjects with high functioning autism and controls. The mean value of EGF was found to be significantly lower in patients than controls. The pattern of distribution of EGF in general population is not known. (Choose one)

- Mann Whitney U test
- ANOVA
- Multiple regression analysis
- Factor analysis
- Fisher exact probability test
- Chi square test
- Paired t test
- Unpaired t test
- McNemar test
- Wilcoxon rank sum test
- Spearman's correlation test
- Pearson's test

Explanation: Mean scores from a scale can be compared in the same group before and after intervention using **paired t test**. When comparing proportions in two groups, **Chi Square** is often used. **Fisher exact correction** is a technique required if any of the expected cell values fall below 5. **For most EMIs if you have to choose Chi Square, you can also consider choosing Fisher exact test.** Mann Whitney U test is an unpaired t test for non-parametric distributions. In this question as EGF distribution is unknown, ideally we should not assume parametric conditions.

Different Statistical Tests

	Comparison				Association	Regression
	2 data sets		>2 data sets		Correlation between 2 variables	Predicting one from another
	Paired	Un-paired	Paired	Un-paired		
Normal Distribution (Mean)	Paired t test	Un-paired t test	Repeated measure ANOVA	One way ANOVA	Pearson correlation	Linear regression
Non - Normal Distribution (Median)	Wilcoxon signed-rank test	Wilcoxon rank sum test Mann-Whitney U test	Friedman test	Kruskal-Wallis test	Spears Rank correlation	Non Parametric Regression
Dichotomous data	McNemar's Test	Chi-square test Fisher's exact test	Cochran's Q test	Chi-square test	Contingency Coefficient	Logistic Regression



72. Conditions to use fixed analysis, what to use? **Error is within the study** --is the answer. Error was within and in between the study, no uncertainty, No uncertainty was among options (heterogeneity was not among options)

[SPMM] In which of the following situations a random effects analysis is indicated in a meta-analysis?
Select one:

- A. Presence of clinical heterogeneity
- B. Presence of too few studies for efficient combining
- C. Presence of statistical heterogeneity**
- D. Meta-analysis of studies other than RCTs.
- E. Presence of methodological heterogeneity

When combining the outcomes from different studies, one may use a fixed and/or random effects model. A fixed effects model assumes that all the studies share the same common treatment effect while a random effects model assumes that they do not share the same common treatment effect. In fixed effect analysis the inference is restricted to included set of studies. It assumes that only random error within studies could explain observed differences. It ignores between-study variations (hence heterogeneity). So this can be applied only if heterogeneity can be safely excluded by testing for it. Random effects analysis assumes that each study shows a different effect which are normally distributed around true mean. This assumption gives proportionally greater weight to smaller studies. Hence this model is susceptible to publication bias and results in wider less precise confidence intervals.

[SPMM] What is the name of the statistical approach preferred to conduct meta-analysis when significant heterogeneity exists?
Select one:

- A. Sampling distribution of the mean
- B. Random effects model**
- C. Unimodal distribution
- D. Fixed effects model
- E. Two-tailed test

Random effects model is used in a meta-analysis when there is evidence of statistical heterogeneity. In this approach, the observed effects between-study variability is incorporated into the meta-analysis calculation prior to calculating a pooled overall effect estimate. The fixed effects model approach is used when heterogeneity has been ruled out. The summary of results obtained is straightforward which will give a pooled overall effect estimate.

[SPMM] Which of the following is the most important assumption made when choosing fixed effect analysis in a systematic review
Select one:

- A. Difference in outcome of individual studies is due to geographical differences
- B. No difference is noted between the outcome of individual studies
- C. Difference in outcome of individual studies is due to different scales used
- D. Difference in outcome of individual studies is due to chance**
- E. Difference in outcome of individual studies is due to variations between studies

The usual method of meta-analysis assumes that all the studies are estimating the same effect and only random variation between research subjects within a study leads the observed study effects to vary. We call this the fixed effects model. Here we assume that the effect to be estimated is the same in all studies and use only the sampling variation within the studies. In other words, we assume that any difference between studies is due to chance. When one cannot explain why different studies exploring the same phenomenon throw up different results and estimates, we can use the random effects model to arrive at pooled estimate. This means we are accepting the existence of the heterogeneity and allow for the degree of uncertainty that results while calculating summary effects. When using a random effects model, we regard each study as estimating a different effect in a population. We estimate both the variability between subjects within the studies (consider this to be equivalent to sampling variation) and the variation between studies (population variability). The other stated differences pertain to methodological variability.

73. Mean is 100, sd 25, population is also 25....where will be 2 sd. Answer – 25 divided by square root answer 25.. 10 on either way .. Answer would be 90.2 to 109.8 as per the options

They have obviously used standard error instead of standard deviation to calc. Hence the answer. $SE = 5$. $2SE$ is $X \pm 1.96SE$ which then comes is to 9.8 when you add and subtract that it gives the answer of 90.2 and 109.8

SE= SD of the mean

[Psych-Mentor] A study looking at glucose levels in a population of people taking antipsychotics finds a mean value of 7mmol/L. The standard deviation of the sample is 6mmol/L and the sample size is 9. The standard error of the mean is 2mmol/L. Which of the following is the closest approximation for the correct range for the 95% confidence interval?

- A. -5-19mmol/L
- B. 1-13 mmol/L
- C. 5-9 mmol/L
- D. 1-7mmol/L
- E. 3-11 mmol/L**

The 95% confidence interval is calculated by the mean $\pm 1.96 \times SE$ (standard error), this is commonly simplified as mean $\pm 2 \times SE$.

Therefore $\pm 2 SE$ of the mean equals 3-11mmol/L.

5-9 mmol/L represents the 68% confidence interval ($\pm 1 \times SE$).

1-13 mmol/L represents the 99.7% confidence interval ($\pm 3 \times SE$)

Confidence intervals are calculated from standard errors NOT standard deviation (as is often stated). Be careful not to confuse the two terms.

74. Bipolar depression in kids, how to treat? **Fluoxetine and olanzapine**

Which one of the following is a true statement about bipolar disorder in children and adolescents?
Select one:

- A. 60-70% of initial presentations are a manic episode
- B. Males predominate in adolescent cases
- C. Manic children typically present with atypical or mixed features**

- D. Lamotrigine is used if treatment resistant manic episodes are present
- E. The course is often episodic and of brief duration

Explanation: Children experiencing a manic episode typically present with atypical or mixed features characterised by irritability, labile mood, and behavioural problems. Periods of depression are common, and early-onset bipolar disorder more often presents with depression than in adult-onset bipolar. Early-onset bipolar may be characterised by non-episodic, chronic, ultra-rapid cycling, mixed irritable and manic states. The prevalence in adolescents is approximately 1% and as high as 6% when including soft subsyndromal symptoms. M>F in childhood cases; M=F in adolescents. For the treatment of mania and hypomania in youth, NICE guidelines suggest following the same recommendations as adults: second-generation antipsychotics (SGAs) may be used as first-line treatment, and mood stabilisers can be added after failure of two trials of SGAs. Lamotrigine has not been studied in RCTs for the treatment of acute bipolar depression in children and adolescents and is not recommended as first-line. It is not recommended in the treatment of manic episodes.

Specific issues

- Bipolar depression is a common clinical challenge and its treatment has been studied much less in youth compared than in adults (see Table 5.2). Antidepressants should be used with care and only in presence of an antimanic agent.⁵ There is limited evidence for the benefit of antidepressants in bipolar depression in adults.⁹ Because of the dearth of trials in youth, we are compelled to extrapolate from adult studies⁵ and recommend use of the olanzapine/fluoxetine combination or quetiapine as first-line treatment, along with lurasidone which is supported by evidence from trials in children aged 10–17.^{10–12}
- The exact relationship between ADHD and BD is still debated. Some evidence suggests that stimulants in children with ADHD and manic symptoms may be well tolerated¹³ and that they may be safe and effective to use after mood stabilisation.¹³ Caution and experience with prescribing these drugs are required (see Tables 5.3 and 5.4).
- The DSM-5 has introduced the new category of Disruptive Mood Dysregulation Disorder (DMDD) to capture severely irritable children (who were commonly misdiagnosed as having BD in the United States). There is as yet no established treatment for DMDD; lithium is ineffective,¹⁴ but SSRIs and psychological treatment options, such as parenting interventions, may be considered.¹⁵

Treatments in bipolar disorder

Choose one best management option for each question below:

A 44-year-old woman presents with a relapse after 3 episodes of mania. Presently she has a mixed episode with both manic and depressive features. She is currently medication free as she developed renal side effects with lithium in the past.

A 17 year old boy diagnosed with bipolar illness by tertiary services presents with first episode of mania. Choose a drug licensed for this indication.

A bipolar patient with treatment resistant depression not responding to quetiapine, olanzapine and antidepressant augmentations. She is already on valproate as mood stabiliser.

A 49-year-old man has a history of bipolar disorder. He is not on any medication for the last 2 years. He is now presenting with symptoms of mania. Choose one drug useful for treatment and prophylaxis of bipolar disorder

Choose...

Lithium
Risperidone
Sodium valproate
No treatment
Gabapentin
Carbamazepine
SSRIs
Olanzapine
Lamotrigine.
ECT
Quetiapine

Your answer is incorrect.

Explanation:

Case 1: Sodium valproate should be used in mixed episodes of bipolar disorder and is a good alternative choice for lithium, in patients with renal impairment. Mixed episodes should be treated as a manic episode and the best evidence for efficacy in treating mixed episodes is for sodium valproate, which has been shown to be more effective than lithium or placebo (Mitchell et al 2004).

Case 2: Lithium monotherapy is the first line in this scenario. It is effective in both manic and depressive relapses, although it is more effective in preventing mania. Lithium is indicated and licensed in the treatment of moderate to severe mania with a NNT of 6.

Case 3: Lamotrigine appears to be effective both as a treatment of **bipolar depression** and as prophylaxis against further episodes. It does not induce switching or rapid cycling.

Case 4: Lithium is useful for both the treatment and prophylaxis of bipolar disorder. It reduces the number and severity of relapses. NICE supports the use of lithium as a first line mood stabilizer.

The correct answer is: A 44-year-old woman presents with a relapse after 3 episodes of mania. Presently she has a mixed episode with both manic and depressive features. She is currently medication free as she developed renal side effects with lithium in the past. → Sodium valproate, A 17 year old boy diagnosed with bipolar illness by tertiary services presents with first episode of mania. Choose a drug licensed for this indication. → Lithium, A bipolar patient with treatment resistant depression not responding to quetiapine, olanzapine and antidepressant augmentations. She is already on valproate as mood stabiliser. → Lamotrigine., A 49-year-old man has a history of bipolar disorder. He is not on any medication for the last 2 years. He is now presenting with symptoms of mania. Choose one drug useful for treatment and prophylaxis of bipolar disorder → Lithium

75. Donepezil 10mg with ses what to do? **Reduce the dose**

[SPMM] An 80-year-old woman was recently started on donepezil 10mg. She presents with gastrointestinal side effects. What is the most appropriate next step in managing her case?
Select one:

A. Reduce the dose of donepezil

- B. Switch to galantamine
- C. Switch to rivastigmine
- D. Switch to memantine
- E. Add pantoprazole

When adverse effects occur with acetylcholinesterase inhibitors, they are largely predictable: excess cholinergic stimulation leads to nausea, vomiting, dizziness, insomnia, and diarrhoea. Such effects are most likely to occur at the start of therapy or when the dose is increased. They are dose-related and tend to be transient. The recommended starting dose for donepezil is 5mg daily, so it would be reasonable in this case to reduce the dose to 5mg and continue to monitor her condition. GI effects appear to be more common with oral rivastigmine than with other cholinesterase inhibitors.

[SPMM] What side effect would you monitor closely in a patient taking Donepezil?

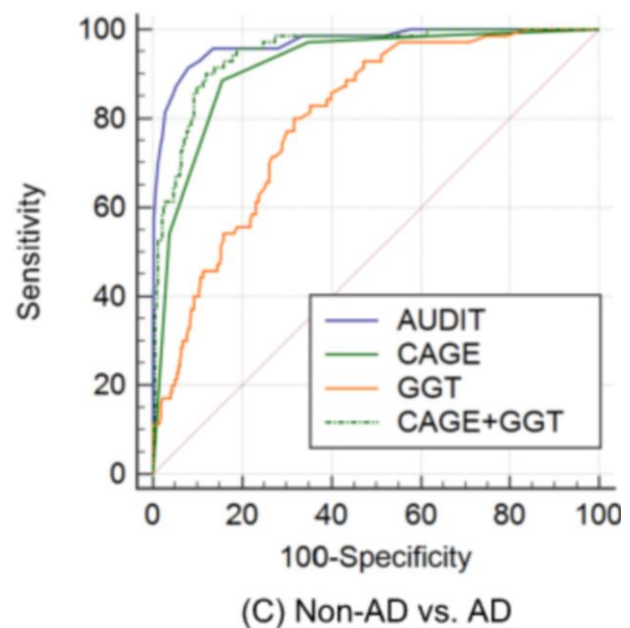
Select one:

- A. Seizures
- B. Tremors
- C. Elevated Blood pressure
- D. Reduced Pulse rate**
- E. Rash

Acetyl cholinesterase inhibitors (AChEs) like Donepezil may have vagotonic effects on heart rate leading to bradycardia. AChE inhibitors can induce sinus bradycardia, sino-atrial block, and aggravate pre-existing sinus node disease and atrioventricular block (Savie et al, 1998). Donepezil overdose has been reported to result in profound sinus bradycardia that can be reversed with atropine

76. ROC best test in primary care had best specificity and sensitivity for relapse

- A. **AUDIT** *this one for primary care*
- B. CAGE
- C. GGT
- D. CDT - *very good one , very specific but not in practice cause expensive*



77. What drug improves bipolar depression lamotrigine it is 3rd line and prophylaxis quetiapine , this one for this question

1. Fluoxetine + Olanzapine
2. Quetiapine only
3. Olanzapine only
4. Lamotrigine only / Lurasidone only
5. Lithium only

78. The BALANCE trial. **See Q. 42**



Red: High impending risk to life



Amber: Alert to high concern for impending risk to life



Green: Low impending risk to life

Medical history and examination

Weight loss	Recent loss of weight of ≥ 1 kg/week for 2 weeks (consecutive) in an undernourished patient ³⁴ Rapid weight loss at any weight, e.g. in obesity or ARFID	Recent loss of weight of 500–999g/week for 2 consecutive weeks in an undernourished patient ³⁶	Recent weight loss of <500 g/week or fluctuating weight
BMI and weight	- Under 18 years: m%BMI³⁵ $<70\%$ - Over 18: BMI <13	- Under 18: m%BMI 70–80% - Over 18: BMI 13–14.9	- Under 18: m%BMI $>80\%$³⁶ - Over 18: BMI >15
HR (awake)	<40	40–50	>50
Cardio-vascular health^{37, 38}	Standing systolic BP below 0.4th centile for age or less than 90 if 18+, associated with recurrent syncope and postural drop in systolic BP of >20 mmHg or increase in HR of over 30bpm (35bpm in <16 years)	Standing systolic BP <0.4 th centile or <90 if 18+ associated with occasional syncope; postural drop in systolic BP of >15 mmHg or increase in HR of up to 30bpm (35bpm in <16 years)	- Normal standing systolic BP for age and gender with reference to centile charts - Normal orthostatic cardiovascular changes - Normal heart rhythm



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Assessment of hydration status	- Fluid refusal - Severe dehydration (10%): reduced urine output, dry mouth, postural BP drop (see above), decreased skin turgor, sunken eyes, tachypnoea, tachycardia	- Severe fluid restriction - Moderate dehydration (5–10%): reduced urine output, dry mouth, postural BP drop (see above), normal skin turgor, some tachypnoea, some tachycardia, peripheral oedema	- Minimal fluid restriction - No more than mild dehydration ($<5\%$): may have dry mouth or concerns about risk of dehydration with negative fluid balance
Temperature	$<35.5^{\circ}\text{C}$ tympanic or 35.0°C axillary	$<36^{\circ}\text{C}$	$>36^{\circ}\text{C}$
Muscular function³⁹: SUSS Test	Unable to sit up from lying flat, or to get up from squat at all or only by using upper limbs to help (Score 0 or 1)	Unable to sit up or stand from squat without noticeable difficulty (Score 2)	Able to sit up from lying flat and stand from squat with no difficulty (Score 3)
Muscular function: Hand grip strength⁴⁰	Male <30.5 kg, Female <17.5 kg (3rd percentile)	Male <38 kg, Female <23 kg (5th percentile)	Male >38 kg, Female >23 kg
Muscular function: MUAC⁴¹	<18 cm (approx. BMI <13)	18–20cm (approx. BMI <15.5)	>20 cm (approx. BMI >15.5)
Other clinical state	Life-threatening medical condition, e.g. severe haematemesis, acute confusion, severe cognitive slowing, diabetic ketoacidosis, upper gastrointestinal perforation, significant alcohol consumption	Non-life-threatening physical compromise, e.g. mild haematemesis, pressure sores	Evidence of physical compromise, e.g. poor cognitive flexibility, poor concentration



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ECG

abnormalities

<18 years: QTc >460ms (female), 450ms (male) 18+ years: QTc >450ms (females), 430ms (males) - And any other significant ECG abnormality	<18 years: QTc >460ms (female), 450ms (male) 18+ years: QTc >450ms (females), >430ms (males). - And no other ECG anomaly - Taking medication known to prolong QTc interval	<18 years: QTc <460ms (female), 450ms (male) 18+ years: QTc <450ms (females), <430ms (males)
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Biochemical

abnormalities⁴²

- Hypophosphataemia and falling phosphate
- Hypokalaemia (<2.5 mmol/L)
- Hypoalbuminaemia
- Hypoglycaemia (<3 mmol/L)
- Hyponatraemia
- Hypocalcaemia
- Transaminases $>3\times$ normal range
- Inpatients with diabetes mellitus: HbA1C $>10\%$ (86mmol/mol)

Haematology

- Low white cell count
- Haemoglobin <10 g/L

Disordered eating behaviours

Acute food refusal or estimated calorie intake <500 kcal/day for 2+ days